MILITARY SPECIFICATION

VALVES, AUTOMATIC SHUT-OFF, FOR GAS SERVICE

(SIZES 1/4 TO 2 INCHES IPS)

1. SCOPE

1.1 Scope. This specification covers self-contained automatic shut-off valves for gas (air or nitrogen) service.

1.2 Classification. Automatic shut-off valves shall be of the following types, as specified (see 6.2):

Type I - Globe or Y pattern configuration.
Type II - Straight through configuration.

1.3 Pressure ratings. Valves shall have nominal pressure ratings of 400, 1500, 3000, and 6000 pounds per square inch gage (psig) as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
QQ-S-763 - Steel Bars, Shapes, and Forgings, Corrosion-Resisting.

MILITARY
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems Requirements for.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-F-1183 - Fittings, Tube, Cast Bronze, Silver Brazing.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-B-23921 - Bronze, Nickel Aluminum Castings for Seawater Service.
MIL-B-24059 - Bronze, Nickel Aluminum; Rod, Flat Products with Finished Edges, Shapes and Forgings.

STANDARDS

MILITARY
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

MILITARY
810-1385884 - Unions, Fittings and Adapters Butt and Socket Welding 6000 PSI, WOG and OXY (IPS).
810-1385943 - Silver Brazing, 3000 PSI, WOG, IPS for UT Inspection.
810-1385946 - Unions, Bronze, Silver Brazing, WOG for UT Inspection.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York, N.Y. 10001.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B-21-66 - Naval Brass Rod, Bar, and Shapes.
B-61-63 - Steam or Valve Bronze Castings.
B-148-65 - Aluminum Bronze Sand Castings.
B-150-66 - Aluminum Bronze Rod, Bar, and Shapes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first unit inspection. Prior to beginning production a sample shall be examined and tested as specified in 4.2.

3.2 Definitions. The definitions of 3.2.1 through 3.2.9 are applicable to this specification.

3.2.1 Automatic shut-off valve. Automatic shut-off valves covered by this specification shall be sensitive and react only to the pressure differential across the valve. In the open position and under no flow conditions, the valve shall be fully pressure balanced and there shall be no net pressure force tending to open or close the valve, regardless of the static pressure in the valve. The valve shall be actuated to close by the pressure drop across the valve caused by flow. The valve shall be adjusted by adjusting the distance between the poppet and the seat in the open position (i.e. changing the restriction to flow presented by the valve when in the open position) and thereby changing the flow-pressure conditions under which the "trip" pressure drop across the valve occurs. Once shut, the valve shall remain shut until the static pressure differential is reduced as required in 3.5.4.

3.2.2 Set point. The flow-pressure conditions under which the valve will trip shut. Established by the spring pre-load tending to hold the poppet in the open position combined with the flow restriction presented by the valve when in the open position.

3.2.3 Trip differential pressure. The differential pressure across the valve in pounds per square inch (psi) which will cause the valve to trip shut.

3.2.4 Accuracy. The operating band of a given valve. Established by the following points:

(a) The minimum flow-pressure condition at or below which the valve will not trip shut regardless of influences such as spring relaxation, mechanical shock or vibration, etc.
(b) The maximum flow-pressure condition at or above which the valve will not remain open regardless of influences tending to resist closure such as breakloose friction, corrosion, sludge, etc.

3.2.5 Adjustable range. The range of flows (in SCFM at a given inlet pressure) over which the valve can be set to trip shut.

3.2.6 Nominal pressure rating. The approximate maximum pressure which the valve will be subjected to in service under normal conditions.

3.2.7 Design pressure and temperature. The maximum pressure and temperature the valve should be subjected to under any condition. The pressure and temperature upon which the strength of the pressure containing envelope is based.

3.2.8 Proof pressure. The maximum test pressure that the valve is required to withstand without damage. Valve operation is not required during application of the proof pressure, but after the pressure has been removed, the valve shall be capable of meeting all performance requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.

3.2.9 Burst pressure. The maximum test pressure that a valve is required to withstand without complete or dangerous structural failure. The valve is not required to operate either during or following the application of burst pressure. Normally a burst test is not required under this specification.

3.3 Materials of construction. Materials shall be as specified in 3.3.1 through 3.3.5 and shall be suitable for the intended pressures. All materials shall be selected to prevent galling, seizing, or excessive wear between operating parts. This specification of materials is not intended to be restrictive, providing proposed alternate materials will give equal or better service than the materials specified. Naval Ship Engineering Center (NAVSEC) (Auxiliary Equipment Branch) approval is required for the use of any alternate material.

3.3.1 Pressure containing envelope. The pressure containing envelope shall consist of the following basic materials:

(a) 18-8 corrosion-resistant steel - AISI 303, 304, or 316.
(b) Aluminum bronze - grade 9BHT of ASTM B-148, alloy 2 of ASTM B-150, MIL-B-23921, or MIL-B-24059. When castings are used, the body shall be cast as a solid billet. Coring will not be permitted.
(c) Bronze - ASTM B-61.
(d) Naval brass - alloy B of ASTM B-21.

3.3.2 Springs. Springs shall be made of phosphor-bronze, monel, corrosion-resisting steel, or other suitable corrosion-resistant material.

3.3.3 Non-metallic seating inserts, diaphragms, gaskets, O-rings, and similar parts. All non-metallic parts shall be compatible with ships air containing sea water vapor (100 percent humidity) and 2190 TEP lubricating oil.

3.3.4 Valve poppets, seat rings, guide bushings, and other internal trim. All internal trim shall be manufactured from nickel-copper alloy, 303, 303Se, 304, 316 or 17-4PH corrosion-resistant steel, stellite, bronze, or nickel aluminum bronze.

3.3.5 Bolting and other metallic parts not subject to line media. These parts shall be made from materials which provide inherent corrosion resistance in a marine atmosphere without requiring painting, coating, or plating.

3.4 Design and construction requirements.

3.4.1 Design emphasis. Design emphasis shall be placed on simplicity, maintainability, ruggedness, and reliability, as defined in 3.4.1.1 through 3.4.1.4.

3.4.1.1 Simplicity. Valves shall be designed with the fewest separate parts and fewest dynamic seals consistent with satisfactory performance. The valve shall require no adjustments during or after assembly, other than the set point adjustment.

3.4.1.2 Maintainability. Internal parts shall be designed to permit easy disassembly and reassembly with standard tools and to prevent, as far as practical, the incorrect reassembly of parts. Positioning and alignment of all parts in assembly shall employ positive means so that correct reassembly is repeatedly assured. In no case shall parts for a given valve be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.4.1.3 Ruggedness. To the maximum extent practical, valves shall be designed to tolerate and not be easily subject to damage, malfunction, or leakage resulting from foreign particle and other line media contamination or from mishandling. Therefore, valve designs should not incorporate or rely upon excessively fine clearances or finishes. Valves shall be capable of tolerating particulate contamination up to 40 microns in size.

3.4.1.4 Reliability. Valves shall be designed to operate satisfactorily for extended periods of time with minimum need for maintenance, lubrication, or adjustment, when subjected to the conditions normally associated with shipboard service. Valves shall not require maintenance or lubrication at intervals shorter than one year, under continuous use on shipboard applications.

3.4.2 Pressure envelope. The design and test pressures for the pressure containing envelope shall be as specified in table I.

Table I - Design and test pressures.

Nominal pressure rating (psig)	Design pressure (psig)	Design temperature (°F)	Proof pressure (psig)	Burst pressure (psig)
400	480	165	600	1200
1500	1800	165	2250	4500
3000	3600	165	4500	9000
6000	7200	165	9000	18000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.

3.4.3 Maximum allowable trip differential pressure. The maximum allowable trip differential pressure, as defined in 3.2.3, shall be in accordance with table II.

Table II - Maximum allowable trip differential pressures

Nominal pressure rating (psig)	Maximum allowable trip differential pressure (psi)
400	15
1500	40
3000	60
6000	75

3.4.4 Body configuration. All valves shall have inline inlet and outlet ports. Type I valves shall be of basic globe or Y pattern configuration and all internal trim shall be readily replaceable without requiring removal of the valve body from the line. Type II valves shall be of straight through configuration and the valves may be of a design requiring removal from the line for replacement of internal trim.

3.4.5 Body construction. All pressure lines, including the reset bleed line, shall be internally ported. The bonnet and bottom cap (where applicable) shall be attached to the body with bolted flanges or threaded union connections. Bearing surfaces of nuts and bolts, and their respective surfaces on the valve, shall be finish machined.

3.4.6 Interchangeability. All parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation and performance without requiring selection or fitting.

3.4.7 Spring design. Springs shall be designed so that they will not be compressed solid during any operation of the valve. Spring ends shall be squared and ground. When removed from the valve and compressed solid, the spring shall not exhibit a permanent set exceeding 0.010 inches per inch of spring length, measured ten minutes after release of the spring.

3.4.8 Threads. All threads shall conform to Handbook H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used. The design shall be such that standard wrenches can be used on all external bolting.

3.4.9 Set point adjustment. Means shall be provided for adjusting the set point at which flow shutoff occurs through the range specified in 3.5.3. Type I valves shall be adjustable with the valve in the line under pressure. Type II valves may be removed from the line for adjustment. The adjusting device shall be safeguarded against accidental change in set point.

3.4.10 End connections. Both the inlet and outlet connections shall be suitable for the maximum inlet pressure and shall be selected as shown in table III. Union threadpieces shall be cast or forged integral with the valve body or secured to the body with a straight thread and O-ring seal. Pipe threads shall not be permitted.

Table III - End connections.

Pressure rating (psig)	Applicable document
400	MIL-F-1183
1500	810-1385946
3000	810-1385943
6000	810-1385884

Only the pertinent dimensions of the documents listed in table III apply. The materials for the union tailpieces shall be as specified (see 6.2). The materials for the union threadpieces and union nuts shall be selected by the manufacturer, from the materials specified herein, to prevent galling. The following materials, in addition to those listed in the documents in table III may be used for the union threadpieces and union nuts:

(a) H-Monel (QQ-N-288, composition B)
(b) CRES (QQ-S-763, composition 303Se or 416Se)

3.4.11 Manual reset. Valves shall incorporate a manual reset device which is operable with the valve in the line under pressure and which permits equalizing pressure across the poppet. A small continuous bleed bypassing the manual reset and providing automatic reset shall also be incorporated when specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.

3.5 Performance requirements.

3.5.1 Accuracy. The accuracy, as defined in 3.2.4, shall be plus or minus 10 percent of the set point unless otherwise specified (see 6.2).

3.5.2 Set point. The nominal required set point shall be as specified (see 6.2). These valves are designed to respond only to the pressure differential created by the flow through the valve, however, since with compressible flow of air or nitrogen the pressure drop through a given restriction is a function of both the flow and the inlet pressure (see 3.2.2) these valves can be set to shutoff at a given flow only at a specific inlet pressure. Therefore, it is the responsibility of the ordering activity to specify the nominal set point in terms of an inlet pressure (psig) and a flow rate (SCFM).

3.5.3 Range of adjustment. Valves shall be capable of meeting all performance requirements specified in 3.5 when set at any point within plus or minus 25 percent of the nominal specified set point, unless otherwise specified (see 6.2).

3.5.4 Reset pressure. Once shut, the valve shall remain shut until the static pressure differential across the valve is reduced to 105 percent or less of the flow pressure drop at which the valve tripped shut.

3.5.5 Set tightness. The maximum permissible internal leakage (from inlet to outlet) with the valve in the shut position, shall not exceed 60 standard cubic inches per hour (SCIH) per inch of seat diameter. The above does not apply for valves which are specified to include an automatic reset (see 6.2).

3.5.6 External leakage. There shall be no visible external leakage, over a 5 minute period, using either bubble fluid or submerging the valve in water.

3.5.7 Ambient conditions. The valve shall operate satisfactorily and meet the performance requirements specified when subjected to environmental vibration, as defined by type I of MIL-STD-167, mechanical shock, as defined by grade A, class I of MIL-S-901, and the varying ambient pressure and temperature conditions specified (see 6.2).

3.6 Marking.

3.6.1 Body markings. Valve bodies shall have the manufacturer's name or trademark, and flow arrow or "inlet" and "outlet" cast, forged or stamped with round bottom dies on them.

3.6.2 Identification plates. An identification plate of corrosion resisting steel, brass or anodized aluminum shall be securely attached to the valve and shall list the following data or a space therefor:

(a) Manufacturer's name or trademark.
(b) MIL-V-24394, type, and pressure rating.
(c) Nominal set point of valve.
(d) Adjustable range.
(e) Manufacturer's drawing numbers (assembly and certified data (CD)).
(f) Applicable manual number.
(g) Space for 9 digit CID number.

3.7 Drawings. Drawings shall be in accordance with categories A, G or H of MIL-D-1000 as applicable.

3.7.1 Preliminary drawings. Preliminary drawings which are sufficient to permit evaluation of the design and approval of materials, shall be submitted with bids to the procuring activity. These drawings shall show the following:

(a) Accurately scaled sectional assembly which clearly depicts the design and construction of the valve.
(b) Bill of material listing specification, grade, condition and any other data required to fully identify the properties of the materials proposed.
(c) Details of the seat, disc and stem assembly and all other replaceable internal trim.
(d) Outline dimensions, disassembly space, location and size of end connections and mounts.
(e) Estimated weight and limitations on installation.
(f) Reference previous shock and vibration approval for valve and test report numbers.
(g) Recommended assembly torques, or equivalent procedures, for all threaded parts.
(h) Table of spring data.

3.7.2 Final drawings. Final drawings and certification data sheets shall be submitted to the procuring activity for approval within 60 days after date of contract. These drawings shall be in accordance with types II and III of MIL-D-1000/2 except for extent of detail. Only the information required in 3.7.1 need be furnished for the type II drawings. The following data shall be furnished for the type III drawings:

(a) Ship identification.
(b) Applicable assembly drawing number(s).

Source: https://assist.dla.mil -- Downloaded: 202640-19T21:45Z
Check the source to verify that this is the current version before use.

(c) Applicable manual number.
(d) CID (APL) number.
(e) Application description.
(f) Valve description.
(g) The set point and adjustable range of valve.
(h) Deviations in materials or end connections from assembly drawings.

3.7.3 Limited rights legend. When the Government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No._____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

3.8 Manuals. Manuals shall be furnished in accordance with type I of MIL-M-15071. The quantity and distribution of manuals shall be as specified (see 6.2). The following, in addition to that required for type I of MIL-M-15071, shall be included as part of the manual contents:

(a) The approved engineering drawings for the valve (including certification data sheet). These drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, etc., and may eliminate extraneous lines and details to clarify the interaction of parts.
(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.
(c) Instructions to permit overhaul by shipyard or other repair facility. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, etc. Also, the appropriate procedure (that is, part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.
(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.
(e) Adjustment procedures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First unit inspection. The first automatic shut-off valve of a particular size, pressure rating, and adjustable range, furnished under a contract or order shall undergo first unit inspection as outlined in table IV. All subsequent valves of the same size, pressure rating, and adjustable range, furnished shall undergo the quality conformance inspection specified in 4.3. First unit inspection may be waived on subsequent orders for the same valve for the same application.

Source: https://assist.dla.mil -- Downloaded: 202640-19T21:45Z
Check the source to verify that this is the current version before use.

Table IV - First unit and quality conformance examination and test outline

Examination and tests	Test conditions	Acceptance criteria
Visual examination	- - - - - - -	Complete conformance
Proof test	Water, air, or nitrogen at the applicable proof pressure as specified in 3.4.2 applied to: (a) Valve inlet with poppet seated and outlet open to atmosphere. (b) Valve inlet and outlet	No external leakage, permanent distortion, or structural failure.
External leakage	Air or nitrogen at maximum pressure rating applied to inlet and outlet.	As specified in 3.5.6
Seat tightness test (see note 1)	Air or nitrogen at maximum pressure rating applied to inlet with poppet seated.	Leakage to outlet port not to exceed value permitted by 3.5.5.
H.I. shock test (see note 2)	Grade A, class I of MIL-S-901	No structural damage or degradation to performance.
Vibration test (see note 2)	Type I of MIL-STD-167.	No resonant frequency 0-33 cps. No structural damage or degradation to performance.
Operational test (see notes 3 and 4)	Air or nitrogen applied to inlet of valve. Pressure tap at inlet and outlet of valve and flow measuring device installed in line. Inlet pressure maintained at constant value equal to value specified in the contract or order. Establish a flow equal to 90 percent of specified set point. Slowly increase flow to 110 percent of specified set point.	The valve shall not close before 90 percent flow rate is reached. Valve shall have closed before 110 percent flow rate is exceeded. Trip differential pressure not to exceed that permitted in 3.4.3.

NOTES:

1. This test not required for valves provided with automatic reset (see 3.4.11).
2. Shock and vibration tests are not required if the valve has been previously approved for shock and vibration.
3. Where other values of accuracy are specified these shall be used instead (see 3.5.1).
4. The manufacturer shall be permitted the option of establishing the trip differential pressure separately at lower pressure-flow conditions.

4.3 Quality conformance inspection. All automatic shut-off valves shall be subjected to quality conformance inspection. This shall consist of the visual examination, proof test, external leakage test, and operational test outlined in table IV. Any discrepancies or failures shall be corrected prior to acceptance.

4.4 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.

5.1 Domestic shipment and early equipment installation.

5.1.1 Valves.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc with polystyrene (loose-fill) material
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. Automatic shut-off valves specified herein are intended to be installed primarily as safety devices in air or nitrogen systems onboard ship to quickly and automatically shut-off flow under certain excess flow conditions caused by a downstream failure or casualty such as a hose rupture. In order to successfully function as intended, the characteristics of the system and the location of the valve must be such that the minimum pressure drop that can occur across the valve as the result of a casualty will be greater than the maximum pressure drop that can occur across the valve during any normal operating condition. The amount of separation between these points must be sufficient to be compatible with the accuracy (see 3.2.4 and 3.5.1) of the automatic shut-off valve selected. If there is not sufficient separation, or if these points overlap, then either the automatic shut-off valve will inadvertently trip shut during certain normal operating situations or it will fail to trip shut under casualty situations, or both, and therefore successful application of an automatic stop valve will not be possible.

6.1.1 Type I valves are intended to provide maximum maintainability (particularly in systems where removal and reinstallation of the valve body from the line may be difficult because of pipe spring) by permitting valve adjustment and renewal of all valve internals with the valve in the line. Type II valves require removal of the valve from the line for adjustment or repair and are intended for applications where compactness and lightweight are primary considerations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Pressure rating (see 1.3).
(d) Materials for union end tailpiece (see 3.4.10).
(e) Whether automatic reset is required (see 3.4.11 and 3.5.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.

(f) Accuracy required, if other than specified in 3.5.1.
(g) Set point expressed in flow (SCFM) at a specified inlet pressure (see 3.5.2).
(h) Range of adjustment required, if other than specified in 3.5.3.
(i) Range of ambient temperatures and pressures which the valve will be subjected to in service (see 3.5.7).
(j) Preservation, packaging, packing and marking if other than as specified in 5.1 (see 5.2).
(k) When polystyrene "loose-fill" is approved (see 5.3).
(l) Whether for air or nitrogen service.
(m) Maximum normal flow (SCFM) at a specified inlet pressure.
(n) End connection size.
(o) Brief description of application and any special performance or construction requirements.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4820-N208)

☆ U. S. GOVERNMENT PRINTING OFFICE: 1968-393-041/5-[illegible]

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:45Z
Check the source to verify that this is the current version before use.